

AQI Tower — consolidated engineering handoff

D01 / R03M mechanical / E05 control-module fit / 5 October 2026

Decision: ready to hand to the internal engineering team for a build-release review. NOT ready to order all parts, fabricate unchanged drawings or energize. Physical prototypes built: 0.

Use one current design

Four ARCTIC P14 Max ACFAN00287A fans and two proposed IKEA104.633.30 filters in parallel, in the preserved R03M rectangular cabinet. External12V Noctua adapter/controller/hub are the reference power chain. The larger cylindrical outdoor tower is a later project, not this build.

What is now closed digitally

Evidence	What it establishes — and does not establish
Mechanical CAD	493 original assembly objects, STEP and15 drawing files retained. Geometry checks do not approve materials, guards, seals or stability.
E05 control fit	Same300x200x120mm enclosure. Hub rotated/moved, speed controller moved, buttons at X=-15/45/105,Y=55mm. Unchanged assumed rear/cable reservations now clear each other and OEM box geometry. Actual connectors, button stack and supports are unverified.
Software	1,041 host assertions; default Opta target compile/link passed with official core4.6.0. All relay outputs remain disabled. No physical test or protective control implemented.
Pressure and power	At assumed300m3/h total, conditional filter allowance14.316Pa each. Four fans catalogue16.8W; plus Opta max2W gives18.8W before hub/controller/auxiliaries/startup. The24W path limit is not a measured margin.
Mechanical batch	Actual43mm closing gap /36mm sleeve /7mm offset extracted. Robust seal-window solver and four eight-bolt joint-demand groups,10 tests. The broad ASSUMED tolerance case has no valid fixed stop; gasket force/material capacity remain unknown.
Electrical batch	24 power /36 wire-drop /8 input scenarios,8 tests. Fan startup multiplier ceiling1.3095x before auxiliaries; an ASSUMED2A-limited fault cannot establish clearing of a2A fuse. Actual fault/inrush data and protective circuit unselected.
Current parts / tools	28-line current parts register separates exact references/candidates from UNSELECTED items; historicalIH02 retained. One batch-verification command and exact-byte pressure provenance. Visuals/room model remain explanatory, not CFD/CADR proof.

Control decision

START/STOP/RESET are ordinary inputs. STOP is not emergency stop or isolation. RESET does not start fans. Internal speed adjustment is a disconnected-source preset; close the box before operation. No live knob modification proposed. Current OEM Digest19-42 confirms low-power contacts but gives no numerical minimum switching range.

How to read this combined PDF

Pages1–2: current decisions and real release blockers. Pages3–4: E05 layout/input review. Pages5–6: preserved E02 bench-control drawing. Pages7–56: earlier50-page engineering handoff, retained as historical evidence. Where its fit/control or board-build status differs, the E05 check record and current target-build JSON govern; older wiring is not released.

What must happen before construction

These are missing engineering inputs and physical acceptance steps, not items that can be marked complete by generating another document. No build approval has been given.

Five remaining release gates

Gate	Exact deliverable needed / responsible scope
1. Guards / structure	Mechanical reviewer and fabricator: guard reach/strength and seams, material grades, joints, feet, whole-device mass/stability and handling. Revised signed detail drawings and unpowered inspection evidence.
2. Filters / harness	Exact delivered filter tolerance and gasket/compression evidence; fan/extension lead, connector and clamp geometry. Resolve service retention, bypass seal and protected entries. OEM enquiries already sent; replies pending.
3. Rated electrical design	Electrical reviewer: hazard-to-protective-function allocation including restart/recovery; complete selected protection/isolation/switching, startup/fault coordination, actual gauges/terminals/glands, metal-panel treatment, mounts and thermal assessment. Opta software cannot replace this.
4. Actual assembly / commissioning	After revision-specific release and authorized procurement: supplier delivery, fabrication, dry-fit inspection, safe qualified assembly, isolation/stop/restart and abnormal/thermal tests. Signed commissioning precedes performance tests.
5. Performance / acceptance	Test lead: agreed indoor room/targets, calibrated or paid/rented instruments. Record airflow, pressure, bypass, noise, power, temperatures and controlled PM decay/background. No proven CADR or outdoor bubble radius yet.

No-lab construction route

Use a local cabinet/metal fabricator and qualified electrical assembly/test service in Pune, Hyderabad or Mumbai when chosen. College/NGO funding supports those services and approved parts; no college workshop is assumed. Prepare quotations from current drawings clearly marked REVIEW ONLY; production follows signed release, not this PDF.

Critical path — no invented dates

Resolve exact component inputs and protective functions → signed mechanical/electrical revision → funding and purchase authorization → delivery/fabrication → unpowered fit and assembly → qualified commissioning → measured performance. Supplier replies and workshop/test capacity control elapsed time; digital compilation alone does not shorten them to a promised date.

Funding and technical sheets

STAKEHOLDER_FUNDING.xlsx is for actual quotations and milestone funding. TECHNICAL_REVIEW.xlsx and CURRENT_RELEASE_GATES.csv preserve decisions and closure evidence. Stock, prices and dates are UNKNOWN until confirmed. Only released line items can enter a purchase/build authorization.

Unchanged boundary: indoor experimental particle demonstrator. No gas-removal, whole-device HEPA, medical, outdoor heat/rain/tamper or public-clean-air coverage claim. No purchases, supplier messages, hardware flashing or energization performed in this continuation.